Low Rank and Sparsity Based Foreign Object Debris Detection with Millimeter Wave Radar

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***Abstract*— This paper investigates the detection of foreign object debris (FOD) on airport runways using off the shelf 77-GHz automotive radar. These radars are characterized by low power transmissions required for detecting spatially large automotive targets such as pedestrians and vehicles. When used for detecting FOD, the detection performance with classical algorithms is poor due to the very low RCS of the targets. Therefore, we propose exploiting the low rank nature of the background clutter and sparse characteristics of the spatially small FOD targets to reconstruct the radar range-angle ambiguity map. When benchmarked with ordered statistic constant false alarm rate and adaptive coherent estimation algorithms, the algorithm demonstrates better detection and lesser false alarm metrics.**

***Keywords*— Automotive radar, foreign object debris (FOD), FMCW, Low-RCS detection, sparsity constraint, ADMM**

I. INTRODUCTION

Detection and identification of foreign object debris (FOD) such as nuts, bolts, metal fragments, stones and tools, on airport runways is a critical safety function. Left undetected, these debris pose a significant risk during takeoff and landing operations, as high-speed aircraft are particularly vulnerable to tire punctures, engine ingestion, and structural damage. Traditional runway inspection methods rely heavily on manual visual inspections, which are time-consuming, labor-intensive, and prone to human error. To overcome these limitations, automated foreign object detection (FOD) systems based on video [1], lidar [2], infrared sensors and hyperspectral imaging [3] are being explored in literature. Millimeter wave radar is particularly effective for this application since, unlike optical systems, the sensor can be used 24/7 and in adverse weather conditions. There are several commercially available FOD radars such as Elva-1 and Navtech Radar, which can detect small debris at long ranges, and Havelson's FOD radar and QinetiQ-Tarsier FOD systems, where the radar is integrated with optical detection. These radar systems have been customized for the FOD application and are characterized by large power transmissions and sizes and huge deployment costs.

In the last two decades there has been a huge increase in the volume of off-the-shelf (OTS) millimeter wave radars in the 76 - 80 GHz frequency band specifically catering the automotive radar market. These monolithic microwave integrated circuit (MMIC) based radars are compact, cheap, and portable. Further, due to the wide bandwidths and high carrier frequency, they enable high resolution range and Doppler sensing. Finally, due to the small electrical size, they are available with large two-dimensional antenna arrays which can be utilized for rapid electronic beam scanning of the radar channel across azimuth and elevation through digital beamforming. Hence, in this work, we explore the possibility of using OTS millimeter wave radar for detecting FOD.

The problem is particularly challenging since OTS millimeter wave automotive radars are designed to detect and image spatially large targets such as pedestrians and vehicles that are typically of radar cross-sections (RCS) greater than 0 dBsm. However, in aircraft runways, these radars must be capable of detecting tiny debris of very low RCS (<-20 dBsm), with low false alarm rates, in the presence of ground clutter arising from the rough concrete runway surface [4]. Further, they must be capable of real-time processing for continuous runway monitoring. Finally, robust detection and classification algorithms have to be developed when large, labeled runway-specific publically available datasets are very scarce.

Several recent works have investigated novel radar deployment and data processing strategies for detecting FOD with OTS millimeter wave radars. In [5] and [6], the authors considered a fixed installation of the FOD radar. In the former, a classical support vector domain description classifier was integrated with particle swarm optimization to distinguish FOD from ground clutter based on power spectrum features. On the other hand, in [6], they substituted the conventional range processing algorithms based on matched filtering with sparsity based algorithms that reduce the effect of strong sidelobes from clutter from suppressing the weaker returns from the target. Unlike these works, the FOD radar is mounted in an unmanned aerial vehicle (UAV) while scanning the runway in [7]. Most recently in [8], the authors replace the real aperture obtained from a mechanically scanning antenna with a synthetic aperture generated with a virtual array realized with a multiple input multiple output (MIMO) radar configuration.

In this article, we consider the range-angle ambiguity plots obtained by processing wideband millimeter wave MIMO radar data for detecting FOD. We propose to exploit the unique nature of the FOD radar returns to improve the overall detection performance. The ground clutter associated with the FOD radar are typically low rank while the target returns from the ultra-small FOD are sparse in the range-angular space. Therefore, we propose to apply both low rank and sparsity penalties on the optimization function to reconstruct

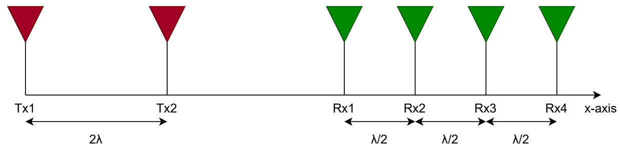


Fig. 1. Virtual linear antenna array formed with two transmitting and four receiving antennas.

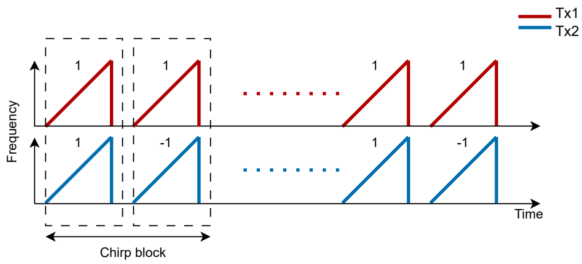


Fig. 2. Binary phase modulated transmission sequence from the two transmitters

the radar ambiguity plot based on [9]. The low RCS targets are subsequently identified from the reconstructed range-angle ambiguity plot through classical constant false alarm rate algorithms (CFAR).

The proposed method is validated with measurement data collected using the Texas Instruments AWR1843BOOST that integrates two transmitting and four receiving antennas to form an eight-element virtual array. The waveform used is a frequency modulated continuous waveform (FMCW) with a radar bandwidth of 2 GHz. We use metal balls of diameter 20mm as isotropic radar targets of -35 dBsm for repeatable measurements. The radar returns are processed to form a range-angular radar signature. We demonstrate an improved detection performance when benchmarked against classical detection strategies: ordered statistics CFAR [10] and adaptive coherence estimation [11].

The article is organized as follows. In the following section, we present the theory including the radar signal model, the signal processing steps and the sparsity based algorithms. This is followed by a short description of the classical radar detection algorithms in Section III. In Section IV, we present the simulation set up and results while Section V discusses the measurement results of the proposed method benchmarked with results obtained from classical detection methods. The final section of the paper concludes the paper.

Notation: We denote scalar variables, vectors, and matrices as a, $\mathbf{a}$, and $\mathbf{A}$ respectively. $\mathcal{C}^{N\times M}$ denotes a complex matrix of N rows and M columns. $(\cdot)^H$ and $(\cdot)^{-1}$ indicate Hermitian transpose and inverse operations on a matrix respectively. () and [] indicate continuous and discrete functions respectively.

II. Radar Signal Processing

A. Radar Signal Model

We consider a MIMO radar with two transmitters (in red) and four receiving antennas (in blue) as shown in Fig.1. The transmitting antennas are spaced 2λ apart while the receiving antennas are uniformly spaced $\lambda/2$ apart along the x-axis where λ is the wavelength corresponding to the carrier frequency. The radar is operated in binary phase modulation (BPM) mode using the coding matrix given by

$$\mathbf{C} = \begin{bmatrix} +1 & +1 \\ +1 & -1 \end{bmatrix}, \tag{1}$$

where, each matrix element, $\mathbf{C}_{l,p}$ indicates the weight applied to the p^{th} transmission pulse from the l^{th} transmitting antenna. Each l^{th} antenna transmits an FMCW signal at a carrier frequency of f_c, with a chirp factor of K, and a pulse repetition interval of T_{PRI} as shown in Fig.2. Therefore the time-domain transmitted signal corresponding to the p^{th} pulse within a pulse pair is

$$\mathbf{s}^{tx}_{l,p}(\tau) = \mathbf{C}_{l,p}\ \mathrm{rect}\left(\frac{\tau - pT_{\mathrm{PRI}}}{T_{\mathrm{PRI}}}\right) e^{j2\pi f_c \tau} e^{j\pi K\tau^2}\ e^{-j4l\pi \sin\theta_d}$$
$$l = 0,1;\ p = 0,1. \tag{2}$$

Here, $rect(.)$ indicates the rectangular signal window of duration T_{PRI}, and τ represents the fast time spanning one T_{PRI}. The transmission along direction θ_d results in the steering component, $e^{-j4l\pi\sin\theta_d}$, from the l^{th} transmitting antenna in (2).

In the FOD radar channel, the targets are typically static. The downconverted and digitized received signal at each m^{th} receiving antenna is

$$\mathbf{s}^{\mathrm{rx}}_{m,p}[\tau] = \sum_{l=0}^{L-1}\sum_{k=0}^{K-1} a_k \mathbf{s}^{tx}_{l,p}\left[\tau - \frac{2r_k}{c}\right]\ e^{-j(4l+m)\pi\sin\theta_d},$$
$$m = 0,1,2,3;\ p = 0,1. \tag{3}$$

where each k^{th} target located at a range r_k from the radar introduces a two-way propagation delay of $\frac{2r_k}{c}$ (c is the speed of light). The two-way path loss incorporates the gains of the radar antennas and the scattering amplitude of the target and is denoted by a_k. $(e^{-jm\pi\sin\theta_d})$ denotes the steering component at the m^{th} receiving antenna array element. The received signal matrix is $\mathbf{S} \in \mathbb{C}^{[M\times P\times N]}$ where M, P and N correspond to the number of receiving antennas, number of pulses within the pulse pair ($P = 2$), and number of fast time samples within one T_{PRI}. Here N is determined by the sampling frequency of the analog to digital converter in the radar receiver channels. The received signal at each m^{th} channel is then decoded using

$$\tilde{\mathbf{S}}_m = \mathbf{C}^{-1}\mathbf{S}_m \tag{4}$$

Note that since the number of transmitting antennas ($L = 2$) is equal to the number of pulses within each transmission ($P = 2$), the BPM coding matrix $\mathbf{C}$ is square and invertible.

The decoded received signal, $\tilde{\mathbf{S}} \in \mathbb{C}^{M\times L\times N}$, is then reshaped to form $\mathbf{Y} \in \mathbb{C}^{ML\times N}$, where the dimension ML indicates the length of the virtual array formed by the L transmitting and M receiving antennas with the steering component $e^{-j(4l+m)\pi\sin\theta_d}$ in (3).

The range profile for each $(m,l)^{th}$ virtual channel is obtained by taking a 1-D discrete Fourier transform along the N fast-time sample dimension (n). Similarly, digital beamforming across the virtual array provides the angular ambiguity diagram. Hence, two dimensional DFT results in the range-angular ambiguity plot given by

$$\mathbf{X}[r,\theta] = \mathcal{F}\left\{\mathbf{Y}[n,(m,l)]\right\}, \tag{5}$$

where r and θ index the range - angle cells. Zero padding or sinc interpolation is carried out during the DFT operation. As a result, the ambiguity diagram is of dimensions $N_r \times N_a$ where N_r and N_a indicate the number of range and angle cells respectively. The above expression shows the range-angle ambiguity plot obtained from a single pulse pair. The process is repeated across multiple pulse pairs corresponding to a single coherent processing interval and the resulting ambiguity plots are coherently integrated to obtain the final range-angle ambiguity plot.

B. Low-Rank and Sparsity Based Method for Enhancing Target Detection

Typically, detection algorithms such as CFAR can be directly applied on the range-angle ambiguity plots generated in the previous subsection. However, in the FOD radar scenario, the radar channel is characterized by very weak target returns due to their low RCS and high ground clutter. Therefore, we implement an additional sparsity based algorithm to separate the target and clutter returns to enhance the detection performance. Essentially, the approach is based on the hypothesis that ground clutter is inherently low rank while returns from the FOD are low in strength but highly sparse due to their extremely limited spatial extent. Therefore, the range-angle ambiguity plot can be considered as

$$\mathbf{Y} = \mathbf{H}\mathbf{X} + \mathbf{N}, \quad \text{s.t.} \quad \mathbf{X} = \mathbf{S} + \mathbf{B}, \tag{6}$$

where $\mathbf{S}$ and $\mathbf{B}$ represents the sparse point target returns and correlated low-rank background clutter respectively, $\mathbf{N}$ is the noise vector, and $\mathbf{H}$ denotes a convolution matrix ($\mathcal{C}^{(ML\times N)\times(N_r\times N_a)}$) along the range-angle dimensions obtained through the two-dimensional Fourier operations along space and fast time. Note that in the above expression in (6), $\mathbf{Y} \in \mathcal{C}^{(ML\times N)\times N_s}$, denotes the raw data collected over N_s PRI and $\mathbf{X} \in \mathcal{C}^{(N_r\times N_a)\times N_s}$ represents the range-ambiguity plots generated for N_s raw data. In this case $\mathbf{H}$ is not a square matrix and its inverse cannot be computed. To make it computationally simpler, an alternative method used in this article is to obtain $\tilde{\mathbf{Y}}$ from the inverse Fourier transform of the $\mathbf{X}$ obtained in (5). This results in a square $\mathbf{H}$ of dimensions $(N_r \times N_a)$. To further reduce the computational time, both $\mathbf{H}$ and $\mathbf{H}^H$ can be pre-calculated and stored.

In order to impose the sparsity penalty, we have to enforce the l_0 norm on $\mathbf{S}$. However, both the minimization of the l_0 norm (number of non-zero terms) and rank minimization are individually NP hard problems. Instead, we relax the sparsity constraint of the target returns to the l_1 norm and the rank minimization to the minimization of the nuclear norm, $||\cdot||_*$, which is the sum of singular values of the matrix as described in [9] and shown in

$$\min_{\mathbf{S},\mathbf{B}} \|\mathbf{Y} - \mathbf{H}(\mathbf{S}+\mathbf{B})\|_2^2 + \mu_1\|\mathbf{S}\|_1 + \mu_2\|\mathbf{B}\|_* \tag{7}$$

Here, μ_1 and μ_2 are the associated regularization parameters. We solve this optimization problem by introducing a Lagrangian multiplier, $\mathbf{Z} = \mathbf{X} - \mathbf{S} - \mathbf{B}$, to transform it into an unconstrained problem, and obtain the objective function as expressed:

$$\begin{aligned} L(\mathbf{S},\mathbf{B},\mathbf{X},\mathbf{Z}) = \|\, \mathbf{Y} - \mathbf{H}\mathbf{X} \,\|_2^2 + \mu_1\|\mathbf{S}\|_1 + \mu_2\|\mathbf{B}\|_* \\ + \langle \mathbf{Z},\ \mathbf{X} - \mathbf{S} - \mathbf{B}\rangle + \frac{\beta}{2}\,\|\mathbf{X} - \mathbf{S} - \mathbf{B}\|_2^2. \end{aligned} \tag{8}$$

Here, $\langle\rangle$ represents inner product, and β is a positive penalty parameter, which controls the convergence speed of the algorithm.

Since the function in (7) involves multiple variables, we use a solution based on Alternating Direction Method of Multipliers (ADMM) where the optimization function in (8) is solved as a series of four smaller problems, involving the four unknown variables, where we minimize the function over one variable while keeping the others fixed. For each i^{th} iteration, we follow the steps below.

1) Update $\mathbf{S}^{i+1}$:

$$\begin{aligned} \mathbf{S}^{i+1} = \arg\min_{\mathbf{S}} \mu_1\|\mathbf{S}\|_1 + \langle \mathbf{Z}^i, \mathbf{X}^i - \mathbf{S} - \mathbf{B}^i\rangle \\ + \frac{\beta}{2}\|\mathbf{X}^i - \mathbf{S} - \mathbf{B}^i\|_2^2 \end{aligned} \tag{9}$$

This problem has a l_1 norm, thus it can be solved using Soft Thresholding algorithm [12].

2) Update $\mathbf{B}^{i+1}$:

$$\begin{aligned} \mathbf{B}^{i+1} = \arg\min_{\mathbf{B}} \mu_2\|\mathbf{B}\|_* + \langle \mathbf{Z}^i, \mathbf{X}^i - \mathbf{S}^{i+1} - \mathbf{B}\rangle \\ + \frac{\beta}{2}\|\mathbf{X}^i - \mathbf{S}^{i+1} - \mathbf{B}\|_2^2 \end{aligned} \tag{10}$$

This subproblem involves nuclear norm minimization and its subgradient computation. Thus it can be solved using singular value decomposition and soft thresholding [13]. The background $\mathbf{B}$ is updated using singular value thresholding on the residual map.

3) Update $\mathbf{X}^{i+1}$: Finally, the subproblem involving the updation of $\mathbf{X}$ is shown below

$$\begin{aligned} \mathbf{X}^{i+1} = \arg\min_{\mathbf{X}} ||\mathbf{Y} - \mathbf{H}\mathbf{X}||_2^2 + \langle \mathbf{Z}^i, \mathbf{X} - \mathbf{S}^{i+1} - \mathbf{B}^{i+1}\rangle \\ + \frac{\beta}{2}||\mathbf{X} - \mathbf{S}^{i+1} - \mathbf{B}^{i+1}||_2^2 \end{aligned} \tag{11}$$

This is a quadratic problem that has an analytical solution given by

$$\mathbf{X}^{i+1} = \left(2\mathbf{H}^H\mathbf{H} + \beta\mathbf{I}\right)^{-1}\left(2\mathbf{H}^H\mathbf{Y} + \beta(\mathbf{B}^{i+1} + \mathbf{S}^{i+1}) - \mathbf{Z}^{i+1}\right) \tag{12}$$

4) Update $\mathbf{Z}^{i+1}$: Finally the Lagrangian multiplier is updated using

$$\mathbf{Z}^{i+1} = \mathbf{Z}^i + \beta(\mathbf{X}^{i+1} - \mathbf{S}^{i+1} - \mathbf{B}^{i+1}) \tag{13}$$

The iterative steps are continued until we reach the stopping condition where the change in $||\mathbf{X}||$ across iterations is very small and below a pre-defined threshold.

III. Standard Detection Algorithms for Benchmarking

A. Ordered statistic CFAR

One of the most commonly used detection strategies on radar signatures is the ordered statistic constant false alarm rate (OS-CFAR) algorithm [10]. Here, for each cell-under-test (CUT), a set of surrounding training cells is collected, sorted, and some n^{th} largest value is chosen as the noise estimate. The CUT is labeled a detection if its magnitude exceeds a scaled version of this ordered statistic. The scaling factor is chosen to meet a desired false-alarm probability.

B. Adaptive Coherence Estimator (ACE)

The ACE is a whitened matched-filter detector that measures through an ACE score how well the received vector in the CUT aligns with the range-angle steering vector corresponding to a hypothesized target [11]. Here, the background noise covariance matrix is estimated from noise only snapshots. A detection is declared when the ACE score of a CUT is above the predefined threshold.

IV. Simulation Results

In this section, we compare the performance of the proposed detection algorithms on simulated radar data. The simulation models the OTS millimeter wave radar signal from Texas Instruments with the parameters presented in Table.1. The radar system utilizes the BPM MIMO scheme implemented via a virtual antenna array with two transmitters and four receivers. The simulation environment replicates a runway scenario containing five static targets of RCS of -35dBsm representing FOD as depicted in Fig.4c at ranges $[2, 4, 6, 8, 10]$ m, all at $0°$ azimuth with respect to the radar array. Additive White Gaussian Noise (AWGN) is added to the target data to model SNR of -10dB. A separate noise-only dataset is generated using the same statistical properties to facilitate covariance matrix estimation. Note that the background noise model, in our simulation, does not model realistic clutter encountered in real world surroundings.

We compare the performance of the three detection algorithms on their success in distinguishing the five FOD targets from the background clutter and noise. The range-angle ambiguity map, shown in Fig. 3a, is derived by processing the radar data as per (5). The FODs at 2 and 4 m can be discerned visually in Fig.3a but the return signal from the FODs at 6, 8, and 10 m are too weak for them to be visually

Table 1. Parameters for TI AWR1843 automotive radar

Parameters	Values
Carrier frequency (f_c)	77 GHz
Sampling Frequency (f_s)	10 MHz
Bandwidth (BW)	2 GHz
Chirp rate (K)	4.5×10^{13} Hz^2
Chirp duration (T_{PRI})	45 μs
Idle time	10 μs
Transmitted power (P_{tx})	14 dBm
Transmitting antennas (l)	2
Receiving antennas (m)	4
Range resolution (m)	0.07

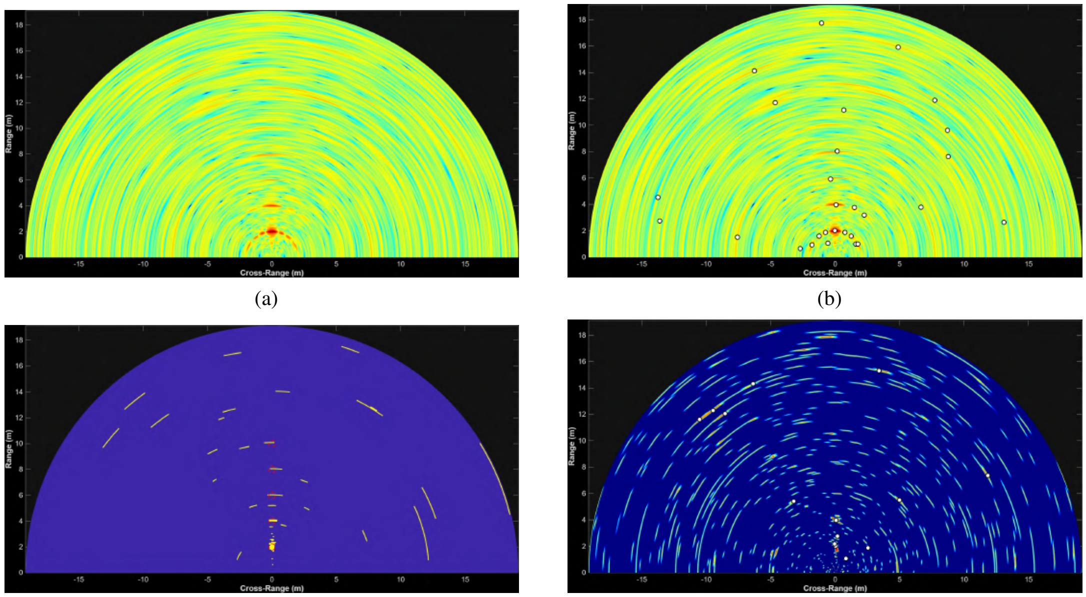
Table 2. Detection and False Alarm Metrics

Data	Algorithm	Detections	False alarms
Simulations	OS-CFAR	4	24
Simulations	ACE	4	142
Simulations	Low rank& sparsity	2	12
Measurements	OS-CFAR	3	0
Measurements	ACE	2	465
Measurements	Low rank& sparsity	4	14

detected. Fig.3b shows the OS-CFAR detections across the RA map. We observe that the algorithm detects the FODs at 2, 4, 6, and 8 m, but we can also see a lot of false alarms. The ACE algorithm whitens the background using the training noise covariance matrix in Fig.3c. The results indicate that ACE detections (yellow gridlines) closely align with the ground truth. However, there are some false alarms in the ranges beyond the furthest simulated FOD at 10 m where the SNR is very low. Next, we implement the proposed low rank and sparsity based ADMM algorithm on the simulated data. The OS-CFAR is applied on the reconstructed range-angle ambiguity plot and the results are presented in Fig.3d. Here, the regularization parameters and penalties were fine tuned to $\mu_1 = 0.1$, $\mu_2 = 10$, and $\beta = 7$. The results show that the FODs at 2 and 4 m are detected, but we also see some false alarms at the range of around 6, 8 and 12 m. The quantitative results listing the detection and false alarms from the three algorithms is presented in Table.2. Here the sparsity based algorithm does not work well possibly since the low rank background clutter was not incorporated in the signal model but utilized in the algorithm.

V. Measurement Results

The measurement set up used for collecting data is shown in Fig.4. Here, TIAWR1843BOOST, a single-chip 76-81 GHz automotive radar shown in Fig.4a is used to capture the the raw digitized radar data directly from the sensor's high-speed Low-Voltage Differential Signaling interface. For this experiment, the radar was mounted at a height of 10 cm above the ground to accurately mimic a runway debris scanning scenario as shown in Fig.4d. The data was captured using the DCA1000EVM raw data capture card and processed offline in MATLAB 2025B. The experimental environment consisted of a paved concrete surface representing an airport runway. The surface roughness of the concrete introduces



(a) (b) (c) (d)

Fig. 3. Simulated-data detection results. (a) Range–angle ambiguity map (b) OS-CFAR (c) ACE (d) low rank and sparsity.

significant ground clutter, which is the primary challenge for FOD detection. Five metallic spheres of diameter 20 mm ($RCS \approx -35$ dBsm), as shown in Fig. 4b, are used as the FOD targets. The targets were placed along the radar boresight ($0°$ azimuth) at distances of 2, 4, 6, 8 and 10 m, as depicted in Fig. 4c. The radar parameters are set as per Table. 1. Note that the simulation and measurement scenarios are made to resemble each other as closely as possible. However, the measurement scenario captures real world effects of clutter from the background including from the ground, the trees and bushes in the vicinity, the direct coupling from the transmitting antennas into the receiving antennas and the mutual coupling between the receiving antennas.

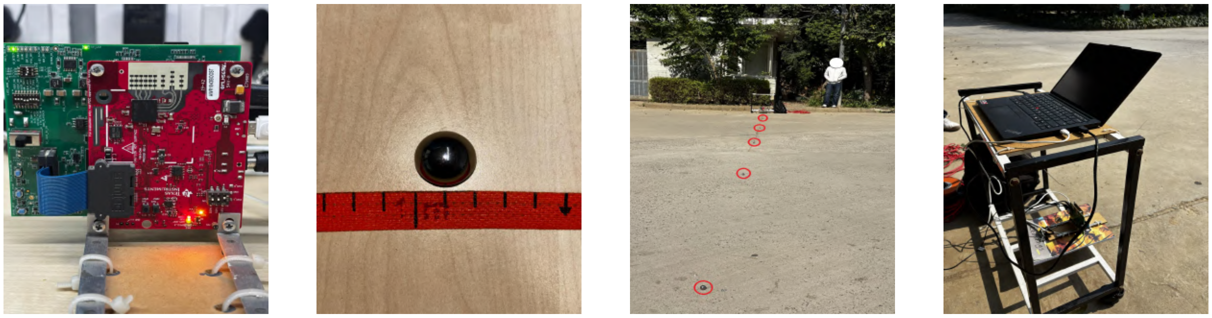
The raw ADC data is processed with the three detection algorithms discussed in the previous section: standard range-angle processing with OS-CFAR, ACE, and the proposed low rank and sparsity ADMM optimization. The standard range-angle ambiguity map obtained via 2D-FFT is shown in Fig. 5a. Due to the presence of background clutter which arises from reflections from the ground, and the bushes away from the radar in Fig.4c, there are several regions in the range-angle ambiguity map which are dominated by clutter and hence it is almost impossible to visually detect the low RCS targets. The OS-CFAR results presented in shown in Fig. 5b indicates the correctly detected FODs at 2, 4, and 6 m. However, FOD targets at 8 and 10 m are not detected due to the weak signal strength. The number of false alarms is also very high.

Fig. 5c presents the detection results from the application of the ACE algorithm. Note that this figure is a binary map where the yellow cells present cells that indicate the presence of target and blue cells indicate the absence of target. In other words, the color bar here does not indicate the strength of the returns. Again, the algorithm detects the nearest FOD but there are several false alarms. The ACE algorithm relies on estimating the sample covariance matrix (SCM),$\hat{\mathbf{R}}$, derived from training data (secondary data). The ACE algorithm was far more effective with the simulation data where the background clutter was modeled as a homogeneous AWGN across the range-angle cells. However, the measurement data exhibits significant non-homogeneity, in the form of rough ground surface texture, small cracks, bushes etc. Further, effects of wind cause the leaves in the bushes and tiny stones on the ground to shake leading to non-stationary clutter. Consequently, more advanced techniques for estimating the whitening filter $\hat{\mathbf{R}}^{-1}$ may be required to effectively nullify the clutter.

The reconstruction results using the proposed ADMM algorithm are presented in Fig. 5d. Compared to ACE detector, the optimization based approach demonstrates better clutter suppression. When integrated with OS-CFAR, We detect targets at 2, 4, 6, and 8 m. Some false alarm clusters are observed around these targets (note the white dots in the results). These detections are not persistent, however, across multiple coherent processing intervals. Hence, when the detection decisions are made in the *track before detect* mode, these false alarms are reduced to enable high detection metrics without high false alarms. The quantitative results presented in Table.2 clearly indicate the superior detection and false alarm metrics of the proposed algorithm compared to the classical detection algorithms. This is mainly because the algorithm handles both the correlated low rank clutter and sparse target returns characteristics.

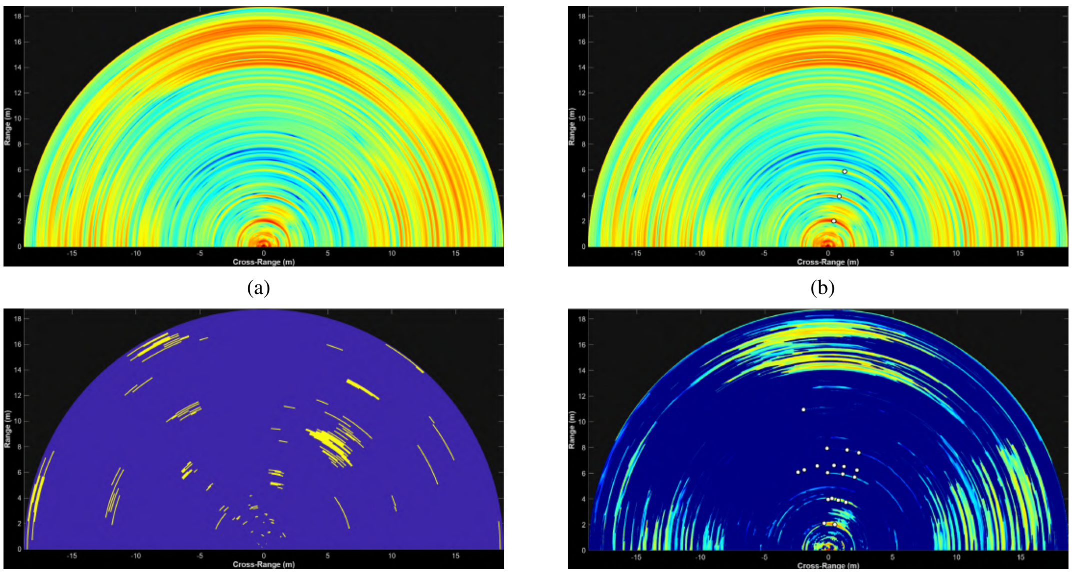
VI. Conclusion

We jointly exploit the low rank of the background clutter and the inherent sparsity of the FOD target returns to improve the detection metrics of very low RCS targets with an OTS



(a) (b) (c) (d)

Fig. 4. Experimental measurement setup. (a) TI AWR1843BOOST FMCW radar (b) Metallic FOD target (20 mm sphere) (c) FODs placed on runway-like ground (d) Radar and laptop setup during measurement.



(a) (b) (c) (d)

Fig. 5. Measured-data detection results. (a) Range–angle ambiguity map (b) OS-CFAR (c) ACE (d) low rank and sparsity.

millimeter wave automotive radar compared to conventional OS-CFAR and ACE. This detection framework is specifically aimed at runway foreign object detection. In our future work, we will explore reducing the computational complexity of the sparsity based architecture to enable real time continuous monitoring of aircraft runways.

References

[1] X. Qunyu, N. Huansheng, and C. Weishi, "Video-based foreign object debris detection," in *2009 IEEE International Workshop on Imaging Systems and Techniques*, 2009, pp. 119–122.

[2] A. Elrayes, M. H. Ali, A. Zakaria, and M. H. Ismail, "Smart airport foreign object debris detection rover using lidar technology," *Internet of Things*, vol. 5, pp. 1–11, 2019.

[3] J. Shan, L. Miccinesi, A. Beni, L. Pagnini, A. Cioncolini, and M. Pieraccini, "A review of foreign object debris detection on airport runways: Sensors and algorithms," *Remote Sensing*, vol. 17, no. 2, p. 225, 2025.

[4] G. Mollo, R. Di Napoli, G. Naviglio, C. Di Chiara, E. Capasso, and G. Alli, "Multifrequency experimental analysis (10 to 77 ghz) on the asphalt reflectivity and rcs of fod targets," *IEEE Geoscience and Remote Sensing Letters*, vol. 14, no. 9, pp. 1441–1443, 2017.

[5] P. Ni, C. Miao, H. Tang, M. Jiang, and W. Wu, "Small foreign object debris detection for millimeter-wave radar based on power spectrum features," *Sensors*, vol. 20, no. 8, p. 2316, 2020.

[6] F. Qin, X. Bu, Z. Zeng, X. Dang, and X. Liang, "Small target detection for fod millimeter-wave radar based on compressed imaging," *IEEE Geoscience and Remote Sensing Letters*, vol. 19, pp. 1–5, 2021.

[7] L. Miccinesi, A. Beni, L. Bigazzi, and M. Pieraccini, "Synthetic aperture radar aboard an unmanned aerial system for detecting foreign object debris on airport runways," *IEEE Access*, vol. 12, pp. 106 735–106 743, 2024.

[8] W. Hong, T. Fan, Y. Zhang, J. Zhou, C. Li, and S. Shichang Gao, "A novel mimo arcsar imaging system for fod detection," *IEEE Transactions on Radar Systems*, vol. 3, pp. 1145–1158, 2025.

[9] W. Li, W. Zhang, Q. Zhang, Y. Zhang, Y. Huang, and J. Yang, "Simultaneous super-resolution and target detection of forward-looking scanning radar via low-rank and sparsity constrained method," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 58, no. 10, pp. 7085–7095, 2020.

[10] S. Blake, "Os-cfar theory for multiple targets and nonuniform clutter," *IEEE transactions on aerospace and electronic systems*, vol. 24, no. 6, pp. 785–790, 1988.

[11] S. Kraut, L. L. Scharf, and R. W. Butler, "The adaptive coherence estimator: A uniformly most-powerful-invariant adaptive detection statistic," *IEEE Transactions on Signal Processing*, vol. 53, no. 2, pp. 427–438, 2005.

[12] E. T. Hale, W. Yin, and Y. Zhang, "Fixed-point continuation for \ell_1-minimization: Methodology and convergence," *SIAM Journal on Optimization*, vol. 19, no. 3, pp. 1107–1130, 2008.

[13] J.-F. Cai, E. J. Candès, and Z. Shen, "A singular value thresholding algorithm for matrix completion," *SIAM Journal on optimization*, vol. 20, no. 4, pp. 1956–1982, 2010.